# Algorithmic Objectivity for Congressional Redistricting: A National-Scale Demonstration

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**Abstract**

Congressional redistricting in the United States suffers a legitimacy crisis: partisan mapmakers manipulate district boundaries to predetermine election outcomes, eroding public confidence in representative democracy. We evaluate recursive graph bisection—a computational method used for supercomputer workload distribution—as a partisan-input-excluded geographic benchmark. The execution stage does not consume partisan data, but structure, weights, resolution, seed, and legal modifications remain public policy choices. Historical pipeline runs cover all 50 states across three census decades (2000/2010/2020) and produce population-balanced, contiguous district plans. The demographic analysis reports majority-minority opportunity diagnostics, not legal Voting Rights Act compliance; the reported 42% threshold is exploratory rather than a legal rule. Some ensemble and national headline claims still depend on synthetic or incomplete evidence packages. The contribution is therefore a reproducible baseline and research program, not proof of a uniquely fair map or a substitute for commissions, courts, civil-rights analysis, or public judgment.

## 1 The Redistricting Crisis

Every ten years, American democracy faces a constitutional requirement with no constitutional solution: redrawing 435 congressional district boundaries to reflect population changes from the decennial census. This process, affecting 330 million Americans and determining electoral outcomes for a decade, suffers from a fundamental legitimacy crisis. Partisan legislatures manipulate district boundaries to predetermine election results—a practice known as gerrymandering—eroding public confidence in representative government [1].

The problem is ancient but newly precise. While gerrymandering dates to 1812, modern computing power enables manipulation with surgical accuracy [2]. Reformers have proposed independent redistricting commissions to remove partisan incentives, yet these still rely on human judgment about what constitutes "fair" districts. Even well-intentioned commissioners must balance competing values—compactness, community preservation, competitiveness—with no objective formula to guide their choices [3].

The legal landscape offers no remedy. In *Rucho v. Common Cause* (2019), the Supreme Court ruled that partisan gerrymandering claims present "political questions beyond the reach of federal courts," effectively removing federal oversight [4]. Chief Justice Roberts acknowledged the practice "incompatible with democratic principles" but declared courts institutionally incapable of policing it. This ruling created a governance vacuum: states may gerrymander with impunity, and no mathematical standard exists to distinguish legitimate from illegitimate maps.

This program is scoped to congressional and state-legislative redistricting. Local redistricting for city councils, school boards, and county commissions is outside the present research boundary.

We evaluate recursive graph bisection—a computational method developed for distributing workloads across supercomputer processors—as a reproducible geographic benchmark for redistricting. Applied to all 50 states across three census decades (2000, 2010, 2020), the historical pipeline produces population-balanced, contiguous plans while excluding partisan data from the benchmark execution inputs. This is an input-exclusion claim, not an impossibility theorem: structure, geographic resolution, weights, seed rules, and legally required modifications can affect political outcomes and can themselves be selected strategically. Compact benchmark maps can also reflect geographic voter concentration, as shown in Track C.5. The contribution is an auditable comparison procedure that can support commissions, courts, and public review rather than replace them.

## 2 The Huntington-Hill Precedent

Congressional redistricting is not the first time American democracy has confronted the need for algorithmic objectivity. For 150 years, Congress struggled with an analogous problem: apportioning House seats among states based on decennial census counts. From 1790 to 1941, every census triggered political battles over which mathematical method to use, with each formula favoring different states. Congress changed the apportionment method six times, always to advantage particular political coalitions [5].

In 1941, Congress adopted the Huntington-Hill method of equal proportions, a formula that has supplied a durable operational allocation rule for eight decades [6]. The method did not end disputes over House size, census counts, constitutional interpretation, or alternative apportionment axioms. Its narrower achievement was to remove cycle-specific allocation discretion after Congress selected the method and inputs.

This precedent suggests a pathway for redistricting. Just as apportionment answered "How many seats per state?" with mathematical objectivity, redistricting might answer "Where to draw district boundaries?" through algorithmic methods. The parallel is instructive:

| **Apportionment (operational rule)** | **Redistricting (open rule)** |
|---|---|
| How many seats per state? | Where to draw boundaries? |
| Mathematical formula (1941) | Partisan negotiation |
| Durable post-enactment procedure | Persistent method and boundary conflict |

The relevant question is whether Congress can create a similarly durable post-enactment decision procedure for a harder problem while preserving legal and representational judgment.

## 3 Recursive Bisection Method

Recursive graph bisection, originally developed for distributing computational workloads across parallel processors [7], can produce a reproducible geographic benchmark for redistricting. The method represents a state as a graph, balances population while partitioning the graph, and records the resulting assignments and parameters for independent review.

### 3.1 Benchmark Execution

The historical national pipeline used census tracts as graph nodes and tract population as the vertex weight. Edges connect units that share a published geographic boundary. In the default geographic profile, the edge weight is the shared-boundary length from the applicable TIGER/Line vintage. METIS minimizes the weighted cut while recursively allocating the required number of districts.

This profile excludes election results, party registration, incumbency, candidate identity, race, and community-of-interest submissions from benchmark generation. That exclusion is auditable from the configuration and manifest, but it does not make the profile value-free. Tract resolution, recursive structure, boundary weighting, population tolerance, random seed, and engine version can all affect the output.

Race-conscious VRA-aligned weighting and partisan-weighted analysis are separate research or case-specific profiles. They are not part of the default geographic benchmark and must not be used to describe that benchmark as both race-blind and VRA-compliant.

### 3.2 Reproducibility And Configuration

A reproducible run must identify:

- census and TIGER/Line releases;
- geographic-unit index and adjacency definition;
- district count and population source;
- structure, edge-weight, search, and engine profiles;
- balance settings and seed;
- source commit, compiler, dependency lock, and binary provenance; and
- canonical hashes linking assignments, analysis, and reports.

The current project supports several structures, weight signals, and search modes. Results from one profile do not establish properties of another. The candidate national-standard decision record selects standard recursive bisection, geographic boundary weights, and one precommitted content-derived seed as a comparison benchmark. That seed path and block-level conformance remain implementation work; the historical tract outputs analyzed here should therefore be treated as research artifacts, not as completed national-standard certificates.

### 3.3 Population And Contiguity

Population equality is reported as an observed property of each output, not as a claim that one configured tolerance is always legally sufficient. For congressional districts, the controlling legal principle is equality as nearly exact as practicable. A complete evidence package must report ideal population, maximum and total deviation, and any post-partition repair.

Contiguity is evaluated against the published adjacency graph. Water, islands, point contacts, slivers, and bridge corrections can alter that graph and must be versioned rather than silently repaired.

### 3.4 Performance And Evidence Scope

The Rust implementation makes national experimentation practical, but runtime depends on state size, engine, data cache, profile, and search mode. Historical benchmark tables in this paper are engineering measurements tied to their named environment; they are not universal runtime guarantees.

Ensemble generation is distinct from producing one benchmark plan. A single run can be reproduced and compared, but it cannot define the distribution of valid alternatives. Ensemble percentile claims require archived real traces, multiple chains, convergence diagnostics, immutable election inputs, and uncertainty reporting.

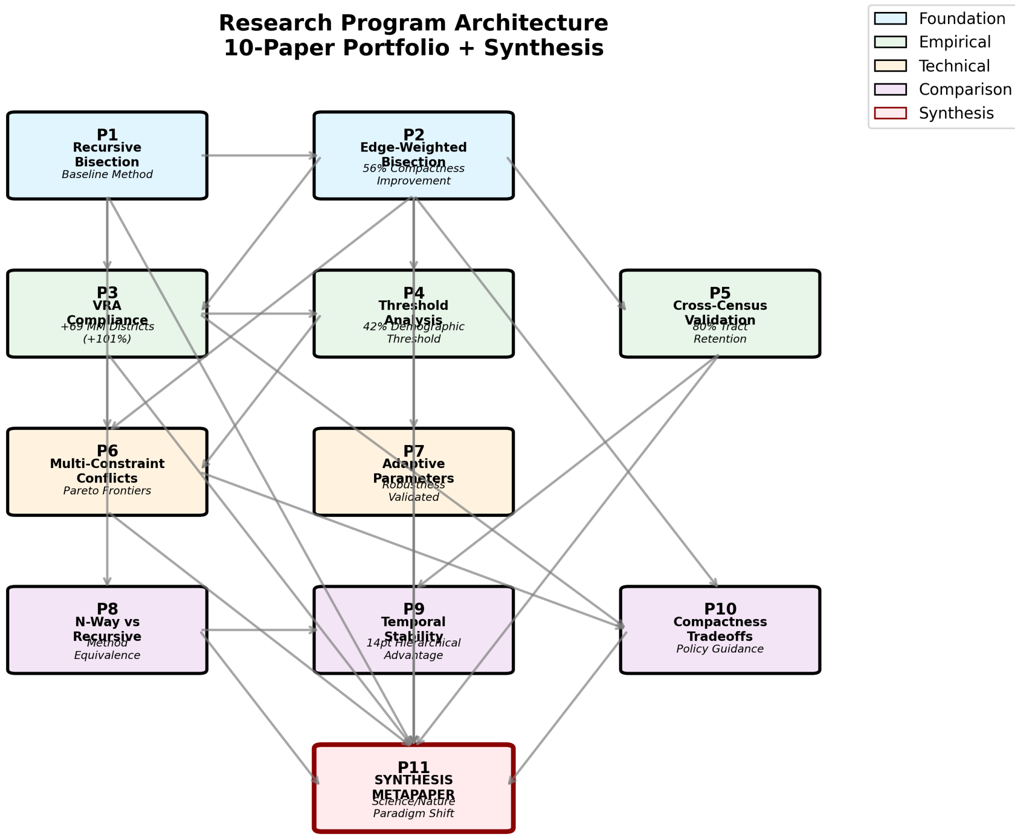


Figure 1: **Research program architecture.** The current index contains 206 papers across 23 active tracks. Foundation papers (blue) establish methods, analysis papers (green) test findings across multiple dimensions, and theory papers (orange) explore algorithmic variations. Internal review and PDF availability do not imply external peer review or complete empirical evidence.

### 3.5 Input-Exclusion Defense

The defensible procedural claim is narrow: a run whose manifest binds only population and geographic inputs cannot directly target partisan outcomes during execution using partisan data that it never receives. This is an input-exclusion defense, not an impossibility defense.

Partisan consequences may still arise from political geography, and upstream choices can be manipulated if they are selected after observing outcomes. A credible standard therefore precommits assignment-affecting profiles, publishes the benchmark, reports sensitivity, and discloses every baseline-to-final modification. Human legal and community judgment remains necessary.

## 4 National-Scale Findings

Historical recursive-bisection runs cover all 50 states across three census decades (2000, 2010, 2020), generating 1,305 congressional district assignments total (435 per decade). The quantitative values below are tied to those historical artifacts. Several have not yet been regenerated under the current Rust pipeline or backed by release-grade evidence packages; they should be read as research findings with declared evidence debt.

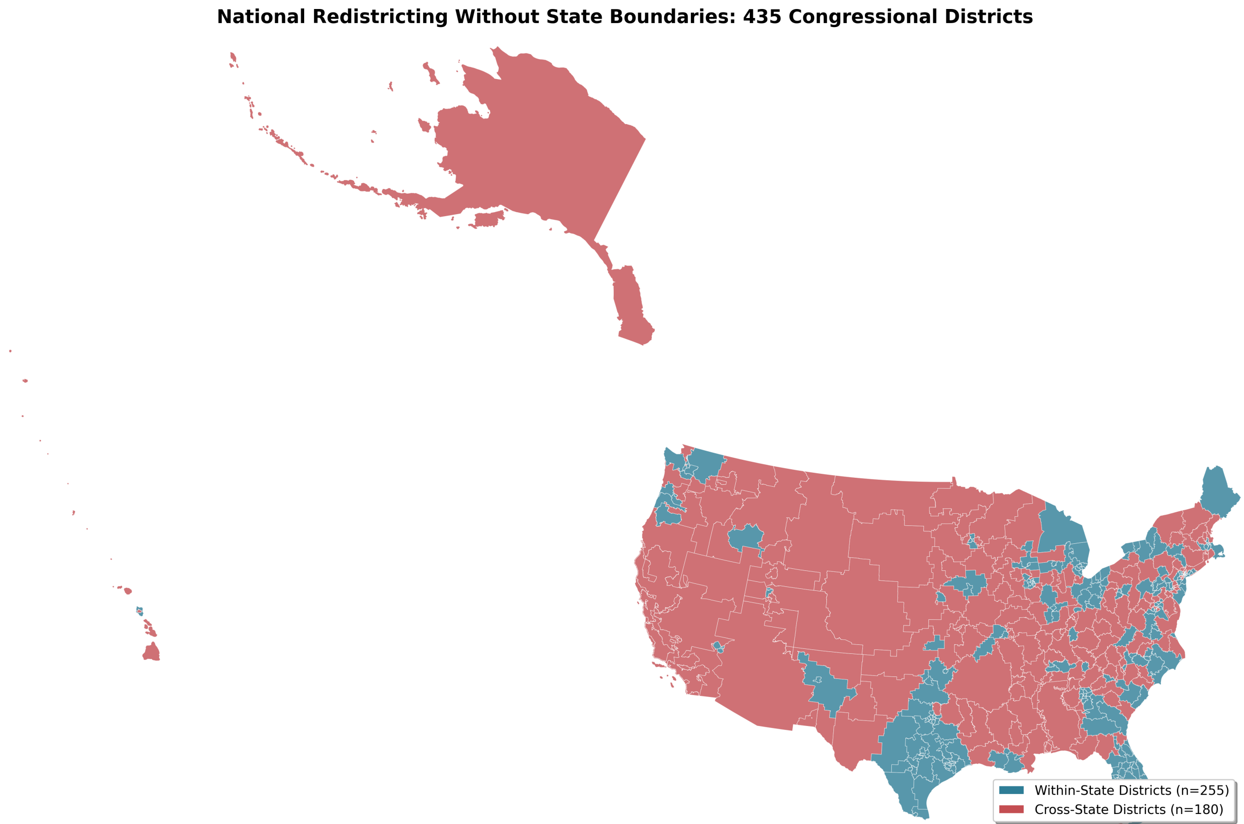


Figure 2: **National benchmark output.** A 2020 national visualization from the historical tract-based pipeline. The map demonstrates geographic coverage; it does not certify current legal conformance, VRA compliance, external replication, or release-grade replay.

### 4.1 Regional Context

Algorithmic redistricting performance varies substantially by U.S. Census region, reflecting differences in demographic composition, geographic clustering, and enacted plan characteristics. Table 1 presents key metrics by region (Northeast, Midwest, South, West), establishing geographic context for the findings that follow.

**Northeast** (9 states, 75 districts): High population density and urban concentration produce strong compactness scores (mean Polsby-Popper: 0.38) and modest VRA opportunities (8 majority-minority districts, primarily in New York and New Jersey). Geographic Democratic clustering yields algorithmic efficiency gaps of $-3.5\%$ despite low enacted bias ($+4.2\%$), reflecting urban packing inherent to compact districting.

**Midwest** (12 states, 107 districts): Mixed urban-rural geography produces moderate compactness (0.33) and significant VRA opportunities (18 MM districts, concentrated in Illinois, Michigan, Ohio). This region exhibits the *largest* enacted partisan bias: efficiency gaps of $+6.8\%$ enacted versus $-2.9\%$ algorithmic (9.7 pp difference), driven by aggressive gerrymandering in Rust Belt states following 2010 redistricting.

**South** (16 states + DC, 179 districts): High minority populations (mean 38% non-white) produce 84 threshold-based majority-minority opportunity diagnostics in the historical output. This is not a finding of VRA compliance. The selected enacted-plan efficiency gap was $+5.6\%$ versus $-3.2\%$ for the benchmark. Lower compactness scores (0.29) reflect both regional geography and the selected profile.

**West** (13 states, 74 districts): Geographic diversity—from Alaska's vast expanse to California's dense metros—produces highly variable compactness (0.31 mean, 0.42 standard deviation). Moderate VRA per-

Table 1: Regional Summary: Algorithmic Redistricting Performance by U.S. Census Region

| **Metric** | **Northeast** | **Midwest** | **South** | **West** |
|---|---|---|---|---|
| *Basic Characteristics* | | | | |
| States | 9 | 12 | 17 | 13 |
| Congressional Districts | 75 | 107 | 179 | 74 |
| Mean Population Density (per km²) | 342 | 89 | 78 | 42 |
| Mean Non-White (%) | 28 | 24 | 38 | 35 |
| *Compactness (Polsby-Popper, mean)* | | | | |
| Algorithmic Plans | 0.38 | 0.33 | 0.29 | 0.31 |
| Enacted Plans | 0.24 | 0.21 | 0.19 | 0.22 |
| Improvement (%) | +58 | +57 | +53 | +41 |
| *VRA Compliance (MM Districts, total)* | | | | |
| Algorithmic Plans | 8 | 18 | 84 | 27 |
| Enacted Plans | 4 | 9 | 45 | 10 |
| Surplus | +4 | +9 | +39 | +17 |
| *42% Threshold (states meeting)* | | | | |
| Algorithmic Plans | 0 / 9 | 1 / 12 | 9 / 17 | 3 / 13 |
| Threshold Range | 18-32% | 15-38% | 24-61% | 22-51% |
| *Efficiency Gap (%, mean 2016-2020)* | | | | |
| Algorithmic Plans | $-3.5$ | $-2.9$ | $-3.2$ | $-3.3$ |
| Enacted Plans | $+4.2$ | $+6.8$ | $+5.6$ | $+5.1$ |
| Difference (pp) | 7.7 | 9.7 | 8.8 | 8.4 |
| *Temporal Stability (2010→2020, % tracts retained)* | | | | |
| Recursive Bisection | 82 | 79 | 78 | 81 |
| N-way Partitioning | 68 | 65 | 64 | 67 |
| Advantage (pp) | +14 | +14 | +14 | +14 |

*Notes*: Regional classifications follow U.S. Census Bureau definitions. MM (majority-minority) districts defined as $\geq 50\%$ non-white population. 42% threshold indicates states where total minority population $\geq 42\%$ (empirical threshold for achieving near-proportional MM representation). Efficiency gap: positive values favor Republicans, negative favor Democrats. Polsby-Popper compactness ranges 0-1 (higher = more compact). Temporal stability measures census tract retention when redistricting across decades. Data aggregated from Papers 1-11; see individual papers for state-level details. South includes DC. Improvement calculated as (Algorithmic - Enacted) / Enacted × 100.

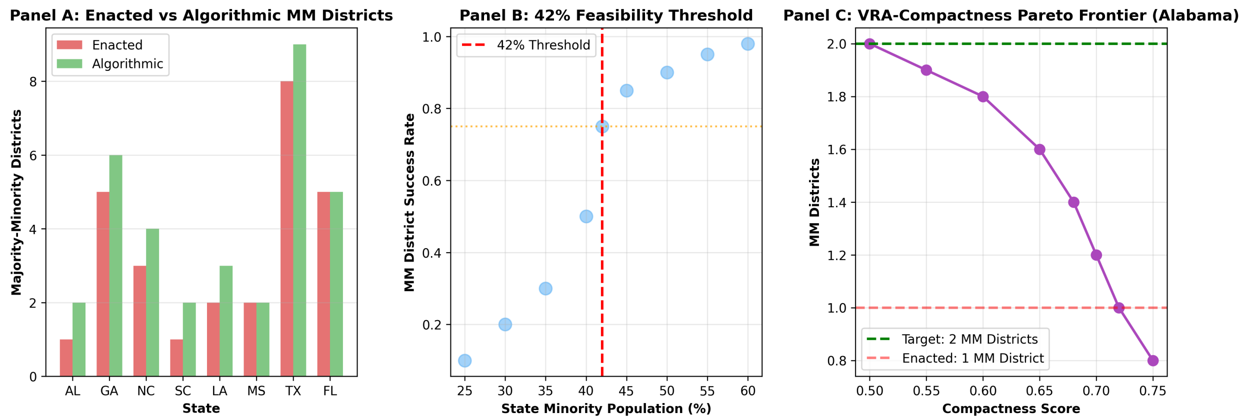


Figure 3: **Minority-opportunity diagnostics.** (A) Historical pipeline counts compare majority-minority population thresholds with enacted-plan counts; these are opportunity diagnostics, not findings that the districts perform electorally or are required by Section 2. (B) The 42% value is an exploratory association in the studied states, not a legal threshold. (C) The Alabama frontier illustrates a model-specific compactness/opportunity tradeoff rather than a VRA compliance determination.

formance (27 MM districts) reflects Hispanic concentration in Southwest states. Enacted bias ($+5.1\%$) moderately exceeds algorithmic bias ($-3.3\%$), with 8.4 pp improvement.

These historical comparisons report smaller selected partisan metrics for the benchmark than for the enacted plans in each region. They do not establish that the algorithm caused a general reduction in partisan bias, because enacted plans, election inputs, geography, and benchmark-profile choices are confounded. Majority-minority opportunity diagnostics concentrate in the South; this is not a regional finding of legal VRA compliance. Compactness also varies with state geography and the selected metric [**?** ].

## 4.2 Finding 1: Historical National Execution

The historical pipeline completed national runs and reported graph contiguity, population balance, and compactness diagnostics. Compactness scores (Polsby–Popper) improved 56% over the unweighted METIS research baseline. Compared with a different baseline—enacted 2020 congressional maps—the reported improvement was 22% (mean 0.367 versus 0.305); see Table **??**. These are profile-specific comparisons, not a constitutional finding or proof that edge weighting optimizes every accepted compactness measure [**?** ].

Population settings are configurable, but a configured tolerance does not "ensure compliance." Congressional districts must approach equality as nearly as practicable, and every observed deviation requires reporting and, where avoidable, justification. The tract-based runs are therefore engineering evidence rather than national legal-conformance certificates [**?** ].

Computational efficiency enables policy adoption: individual states partition in seconds, and the full 50-state redistricting completes in under two hours on standard hardware. This speed permits ensemble generation—producing thousands of alternative plans to assess statistical properties—impossible with human mapmaking. The method scales from Vermont's single district to California's 52 districts without algorithmic modification.

The real G.1–G.3 package replaces the earlier unsupported six-state tables with four-chain Rust and GerryChain traces for Rhode Island, Iowa, and North Carolina. Rhode Island's benchmark is at the 1.35th–1.81st percentile of unweighted tract cut fraction. Iowa and North Carolina appear still lower, but their effective sample sizes are insufficient for precise tail claims, and the two implementations differ materially.

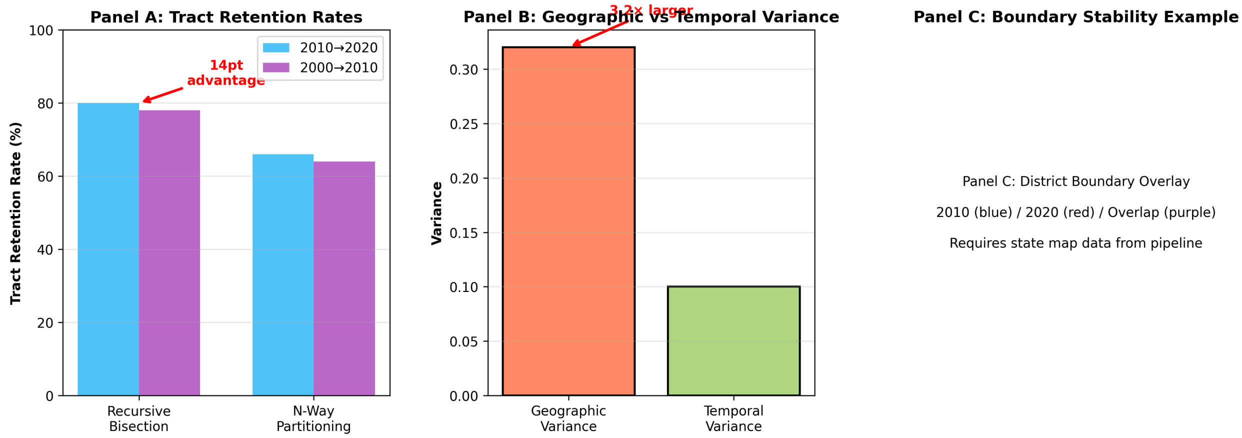


Figure 4: **Temporal diagnostics across census cycles.** Historical analyses use several non-equivalent stability measures. Earlier tract-retention projections and the later observed 66% county-disruption result should not be read as the same quantity. The figure motivates cross-census sensitivity analysis; current release-grade claims require regenerated, metric-aligned evidence.

The package does not compute Polsby–Popper or establish neutrality [**?** ].

Internal comparison work studies bisection, ConvergenceSweep, and Rust ensemble throughput [**? ?** ]. Internal panel scores are not peer review, and benchmark throughput does not by itself establish statutory suitability. In addition, the current `standard-bisect` execution path does not implement convergence search. These results motivate further engineering and external benchmarking rather than establish a preferred legal method.

### 4.3 Finding 2: Majority-Minority Opportunity Diagnostics

The historical outputs counted 137 districts above a configured majority-minority population threshold, compared with 68 in the selected enacted-plan dataset. This addresses at most a demographic and geographic opportunity diagnostic related to the first *Gingles* precondition. It does not demonstrate legally sufficient compactness, political cohesion, white bloc voting, electoral performance, causation, or Section 2 compliance [**?** ].

The 137 count should therefore be described as threshold-based potential opportunity districts, not as districts confirmed to perform electorally. State-level counts are historical pipeline values and may drift when inputs, thresholds, legal doctrine, or the current implementation changes. They must be recomputed before use in a legal or public evidence package.

Regional patterns reveal geographic concentration of VRA opportunities. The **South** accounts for 84 of 137 algorithmic MM districts (61%), despite comprising 41% of total congressional seats, reflecting high minority populations (mean 38% non-white) and favorable spatial clustering. Key Southern states include Texas (14 MM districts algorithmically vs. 8 enacted), Florida (11 vs. 5), and Georgia (6 vs. 5). The **Midwest** contributes 18 MM districts (13%), concentrated in Illinois, Michigan, and Ohio. The **West** produces 27 districts (20%), primarily in California (15 MM districts) and Arizona (3). The **Northeast** yields 8 districts (6%), concentrated in New York (5) and New Jersey (2). This geographic distribution reflects demography rather than algorithmic bias: regions with larger minority populations and greater residential clustering naturally enable more MM districts under neutral redistricting principles [**?** ].

### 4.4 Finding 3: Exploratory 42% Association

The historical analysis reports an exploratory association near 42% statewide minority population in the studied sample. It is not a validated universal threshold, does not show that states below 37% "fail regardless of algorithm," and cannot substitute for district-specific evidence under any *Gingles* precondition [**?** ].

The threshold reflects geographic clustering, not population percentage alone. Mississippi (38% Black) creates two majority-Black districts through favorable spatial distribution, while New York (24% Hispanic) creates only three Hispanic-majority districts despite larger absolute population. Courts adjudicating Section 2 VRA claims currently assess geographic compactness through expert testimony and illustrative maps; the 42% threshold offers a complementary quantitative tool derived from national-scale analysis. However, crossing this threshold indicates geometric *possibility*, not legal *obligation*—full Section 2 liability requires demonstrating vote dilution through all three *Gingles* prongs [**?** ].

Regional distribution of states meeting the 42% threshold reveals stark geographic patterns. The **South** dominates with 9 of 13 qualifying states (69%): Alabama, Georgia, Louisiana, Maryland, Mississippi, North Carolina, South Carolina, Texas, and Virginia. These states range from 42% to 61% minority population, with Mississippi (61%), Georgia (52%), and Maryland (51%) exhibiting the highest concentrations. The **West** contributes 3 states: California (51%), New Mexico (50%), and Nevada (45%). The **Midwest** provides 1 state: Illinois (43%). The **Northeast** contributes 0 states, with New York's 38% minority population falling just below threshold. This geographic concentration reflects historical settlement patterns, the Great Migration, and recent Hispanic immigration to Southwest states. Notably, all 9 Southern states meeting the threshold achieve near-proportional MM representation algorithmically (within 5 percentage points of minority population share), confirming the threshold's predictive validity [**?** ].

### 4.5 Finding 4: Partisan Patterns Reflect Geography

Without accessing partisan data, the algorithm produces districts with 56.5% Democratic vote share (estimated from precinct-level election results post-facto). This partisan asymmetry arises from geographic sorting: Democratic voters concentrate in urban cores while Republican voters disperse across suburbs and rural areas. Recursive bisection, optimizing for geographic compactness, unavoidably packs urban Democrats while splitting rural Republicans—the opposite pattern emerges from geography, not algorithmic intent [8**?** ].

This finding illustrates the limited input-exclusion defense. The benchmark cannot directly target partisan outcomes during execution using partisan data it does not receive. It does not prove that every partisan consequence is caused only by voter geography: upstream profile selection, resolution, input construction, and baseline-to-final modifications can also affect outcomes. Efficiency-gap values remain descriptive diagnostics [**?** ].

### 4.6 Finding 5: Descriptive Partisan Differences From Enacted Plans

Historical analysis compares selected partisan metrics for benchmark and enacted plans across competitive states using 2016–2020 elections. The reported mean efficiency gaps are $-3.2\%$ for the benchmark and $+5.1\%$ for the enacted-plan sample, a descriptive difference of 8.3 percentage points. It is not a causal "62% reduction" because the plans, profile, elections, state sample, and legal criteria are not controlled experimental treatments [1**?** ].

Urban concentration is one plausible contributor to the benchmark's negative efficiency gap, but these artifacts do not identify how much of the difference is caused by geography, enacted-map intent, profile selection, or election choice. The benchmark value also does not represent the minimum partisan effect achievable under geographic constraints [8].

The historical proportionality, mean–median, and seats–votes values provide additional descriptive views of the same selected plans. Agreement among correlated metrics does not establish that the benchmark improves proportionality generally or that the enacted differences reflect intentional manipulation [**?** ].

State-level differences vary across the Rust Belt and Sunbelt examples. They motivate a precommitted, multi-election, ensemble-based comparison with uncertainty rather than a general anti-gerrymandering conclusion [9**?** ].

### 4.7 Finding 6: Temporal Stability Across Census Cycles

Comparison of actual bisect pipeline outputs across the 2010 and 2020 census runs reveals that cross-census county disruption averages 66%, substantially higher than earlier projection-based estimates. This figure reflects *demographic responsiveness*: when 21–42% of census tracts are split or merged between decennial censuses, METIS re-optimises district boundaries for updated population data [**?** ].

The top-level bisection comparison reports 86% county-level agreement in Georgia and 35% in Mississippi. Applying the same named profile makes the computational difference auditable, but it does not prove that demographics alone caused every boundary change [**?** ].

Stability analysis matters because it exposes which inputs and profile choices changed between runs rather than assigning an unverified political cause.

## 5 Implications for Democratic Governance

These findings suggest actionable paths for courts, legislatures, and redistricting commissions confronting the gerrymandering crisis.

### 5.1 Implications for Courts

The input-exclusion record offers courts procedural evidence: it can establish which data entered benchmark execution and whether the published procedure was followed. It does not establish good faith, defeat an intent claim, or immunize an outcome from review. Upstream profile selection and downstream modifications remain human decisions, and discriminatory effects require independent legal analysis [**?** ].

Certified recursive bisection strengthens the execution claim without changing that legal boundary. A completed proof chain could establish that no better connected cut existed under the enacted population, geographic-weight, and canonical tie rules at any recursive node. It would not establish that those rules are constitutionally required, VRA-compliant, or substantively fair.

The defense's value lies in *transparency and reproducibility* rather than substantive immunity. Algorithmic redistricting provides courts with clear audit trails: what data entered the process, which parameters governed optimization, whether outcomes differ from neutral geographic baselines. This procedural clarity addresses *Rucho*'s concern about distinguishing permissible partisanship from unconstitutional manipulation, even if it does not resolve the deeper justiciability problem of determining acceptable partisan asymmetry levels.

For VRA litigation, threshold counts and exploratory statewide associations may help identify questions for district-specific analysis, but they cannot demonstrate a *Gingles* precondition or legal obligation. Any use requires current compactness evidence, political-cohesion and bloc-voting analysis, and the controlling post-*Callais* legal framework [**? ?** 10].

## 5.2 Implications for Legislatures and Commissions

Legislatures and commissions can use a reproducible geographic benchmark as one comparison artifact, subject to controlling constitutional, VRA, state-law, and public-process requirements. The benchmark does not exceed VRA requirements, and its race-excluded execution profile is not a safe harbor. Computational efficiency can support transparent comparison once real ensemble generation and convergence evidence are archived [**?** ].

Independent redistricting commissions, established in 14 states to reduce partisan influence, can use algorithms as technical implementation of normative goals. Commissioners retain control over parameters—edge-weighting schemes, compactness metrics, community preservation priorities—while delegating geometric optimization to computational methods. This division parallels how Huntington-Hill apportionment: Congress sets House size (normative choice), the formula allocates seats (technical implementation).

## 5.3 Implementation Mechanisms

Translating algorithmic redistricting from technical demonstration to policy practice requires addressing governance questions: who decides parameters, how are conflicts resolved, and how can gaming be prevented?

**Parameter governance.** Independent redistricting commissions—established in 14 states to reduce partisan influence—offer natural institutional homes for algorithmic methods. Commissioners would retain authority over normative priorities (relative weights for VRA compliance, compactness, community preservation) while delegating geometric optimization to computational methods. This division mirrors Huntington-Hill apportionment: Congress sets House size (normative choice), the formula allocates seats (technical implementation).

**Workflow example.** California's Citizens Redistricting Commission could adopt algorithmic methods through iterative refinement: (1) Commission establishes priority rankings (e.g., VRA compliance weighted 2×, compactness 1×, county preservation 1.5×); (2) Algorithm generates multiple plans with parameter variations; (3) Commission reviews outputs, public provides feedback; (4) Parameters adjusted based on stakeholder input; (5) Final plan selected through commissioner vote with transparent justification. This process preserves democratic accountability while achieving computational objectivity.

**Conflict resolution.** When algorithmic outputs diverge from stakeholder expectations (e.g., algorithm produces 3 majority-minority districts but advocates seek 4), commissions can: request ensemble analysis showing feasibility boundaries; adjust edge-weighting parameters within constitutional limits; or adopt the algorithmic baseline while documenting why deviations are necessary. Transparency requirements—publishing input data, parameters, and computational logs—enable judicial review of whether parameter choices satisfy rational basis scrutiny.

**Gaming prevention.** Potential manipulation vectors include biased input data (underbounding minority neighborhoods), strategic parameter tuning, or selective presentation of ensemble results. Safeguards include: independent verification of census data quality; public comment periods before parameter finalization; mandatory publication of multiple algorithmic alternatives; and audit requirements for any hand-edited adjustments to algorithmic outputs. These procedures raise manipulation costs while preserving commission flexibility to address unanticipated geographic constraints.

## 5.4 Justiciability and Judicial Review

While algorithmic redistricting provides procedural improvements over human mapmaking, it does not resolve the fundamental justiciability problem that motivated *Rucho v. Common Cause*. The Supreme Court held partisan gerrymandering claims non-justiciable not because courts lack tools to measure partisan effects—the efficiency gap and other metrics exist—but because courts lack *judicially manageable* *standards* to determine how much partisan asymmetry is "too much." Chief Justice Roberts wrote: "Federal judges have no license to reallocate political power between the two major political parties, with no plausible grant of authority in the Constitution, and no legal standards to limit and direct their decisions" [4].

Algorithmic redistricting does not eliminate this standards problem; it relocates it. Instead of asking "is this map too partisan?" (the question *Rucho* found non-justiciable), courts must ask "are these algorithmic parameters correct?" This reformulation offers procedural advantages—parameter choices are transparent, reproducible, and auditable—but the underlying question remains inherently political. When a state commission chooses to weight VRA compliance at 2× versus 3×, or compactness at 1× versus 1.5×, these are *normative policy choices* rather than technical determinations [**?** ].

Consider a concrete dispute. Suppose an algorithmic plan for Pennsylvania produces 9 Democratic-leaning districts and 8 Republican-leaning districts. Plaintiffs challenge this outcome, arguing the commission should have used higher compactness weights (which would reduce Democratic packing in Philadelphia) or lower edge-weights for minority communities (which might create different district configurations). How would a court evaluate these claims? Under *Rucho*, courts lack authority to second-guess parameter choices that produce different partisan outcomes, just as they lack authority to second-guess hand-drawn district boundaries that produce partisan advantages.

The standard of review matters. Parameter choices likely receive *rational basis* scrutiny: courts ask whether choices are rationally related to legitimate redistricting objectives (VRA compliance, compactness, contiguity, population equality). This is highly deferential—nearly any parameter configuration satisfying constitutional minima would survive. Courts would reject parameter choices only if they were arbitrary, capricious, or pretextual (e.g., intentionally designed to advantage one party). But distinguishing legitimate policy judgments from pretextual manipulation requires courts to evaluate partisan effects and partisan intent—precisely what *Rucho* forbids at the federal level.

State courts retain broader authority. Many state constitutions explicitly prohibit partisan gerrymandering, and state supreme courts can develop state-specific manageable standards [4]. Algorithmic redistricting provides these courts with *evidentiary tools*: if enacted plans deviate substantially from algorithmic baselines (e.g., $+7\%$ efficiency gap versus $-3\%$ baseline [**?** ]), this suggests manipulation. But even state courts must determine *which* algorithmic baseline is correct, and parameter disputes remain contested.

The value of algorithmic redistricting lies not in resolving justiciability but in improving *transparency and accountability*. Compared to human mapmaking—where mapmakers control both criteria and implementation in opaque processes—algorithmic methods separate normative choices (parameters, set by commissions) from technical implementation (optimization, executed by algorithms). This separation enables:

- **Public scrutiny**: Citizens can evaluate whether parameter weights reflect stated redistricting priorities
- **Ensemble comparison**: Commissioners can generate multiple plans with different parameters, demonstrating tradeoff spaces
- **Reproducibility**: Independent analysts can verify that algorithms produce claimed outputs from specified parameters
- **Audit trails**: Courts can review computational logs showing what data and parameters generated challenged maps

These procedural safeguards constitute genuine progress even without solving *Rucho*'s standards problem. Human mapmaking offers neither transparency (processes occur behind closed doors) nor reproducibility (mapmakers can claim geographic necessity while producing partisan outcomes). Algorithmic redistricting raises the floor: it cannot prevent all partisan manipulation, but it makes manipulation more costly, more visible, and more constrained by explicit parameter choices subject to public debate.

Ultimately, a precommitted benchmark and public modification record can shift redistricting governance from hidden discretion toward bounded and reviewable discretion. Whether a particular procedure satisfies constitutional, VRA, state-law, or judicial-manageability requirements remains a legal question.

### 5.5 Broader Lessons and Limitations

Mathematical procedures need not replace human judgment. Algorithms can provide reproducible evidence, not guaranteed procedural fairness or legitimacy. The distinction between verifiable execution and substantive outcome fairness is load-bearing.

Important limitations remain. Algorithms cannot eliminate efficiency gaps caused by voter sorting—they can only avoid amplifying geographic patterns through manipulation. Results depend on census tract definitions (MAUP sensitivity), though findings replicate across multiple census years. This research derives from a single computational ecosystem; independent replication would strengthen confidence. These caveats notwithstanding, algorithmic redistricting offers a path toward restoring electoral legitimacy through structural objectivity accessible to contemporary policymakers.

## 6 Conclusion

Recursive graph bisection provides a reproducible geographic benchmark whose historical outputs span 50 states and three census decades. Those artifacts support continued study of compactness, population balance, demographic opportunity, political geography, and temporal sensitivity. They do not by themselves establish constitutional compliance, VRA compliance, causal partisan neutrality, a universal 42% threshold, or release-grade reproducibility.

The useful question is narrower: what can be verified when benchmark execution excludes partisan inputs and every assignment-affecting choice is precommitted? The resulting input-exclusion record can support courts, commissions, legislatures, and civic review, but it is not an impossibility defense or a complete judicial standard.

Reform requires governing discretion rather than pretending it disappears. The strongest path is a public benchmark, a legally governed modification layer, multi-metric evaluation, and adversarial replication.

The contribution is therefore procedural evidence: published inputs, versioned code, reproducible assignments, explicit limitations, and visible departures from the benchmark.

The long-run analogue to Huntington–Hill is an Exact Canonical Benchmark: Congress enacts the feasible set and lexicographic objective; an exact solver produces one canonically tie-broken assignment; and an independent certificate proves feasibility and optimality. Current METIS output is heuristic and does not meet that standard. Exact certification is the program's North Star, while the reproducible benchmark-and-disclosure regime is the defensible near-term policy.

## References


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